



Machine Learning & Computational Biology Industrial Placement 2025/26

Aryan Isaac Bhobe

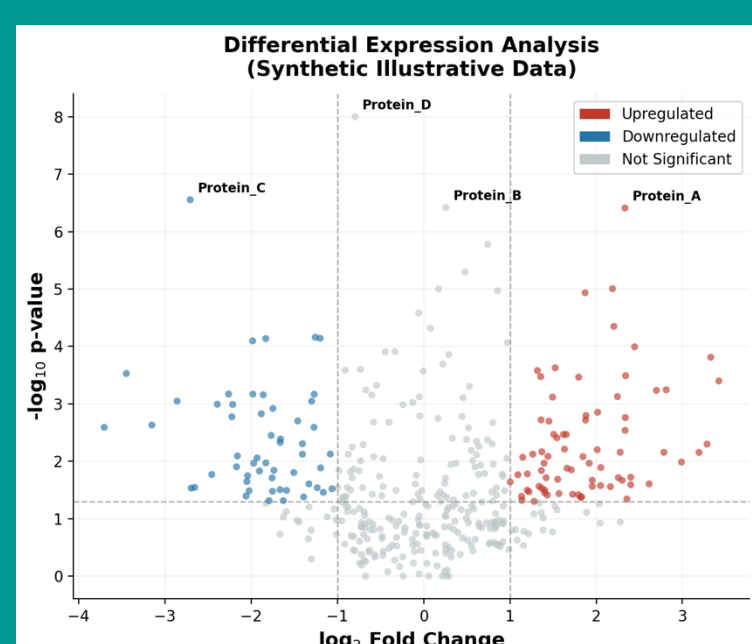
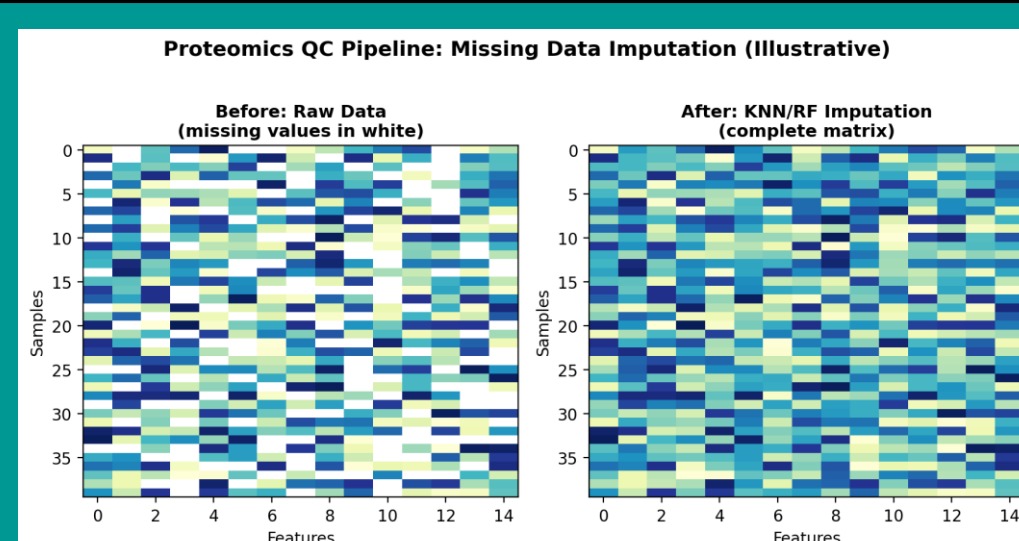
B.Sc (Hons) Computer Science & Artificial Intelligence

ABOUT MERCK SHARP & DOHME (MSD)

- MSD is a research-intensive biopharmaceutical and global health care company. It develops and supplies prescription medicines, vaccines, biologic therapies, and animal health products aimed at preventing and treating a wide range of diseases.
- Merck Research Laboratories (MRL)** is the research arm of MSD that focuses on turning basic science into new therapeutic products, especially in key areas like oncology, infectious diseases, and cardiometabolic conditions.
- Within MRL:**
 - Biology and chemistry teams working on new drug targets and molecules.
 - Quantitative sciences: biostatistics, data science, machine learning, and modelling, supporting trial design, biomarker development, and portfolio decisions.



Merck Sharp & Dohme LONDON
Francis Crick Institute (Left) [June – September]
120 Moorgate (Right) [September – June]

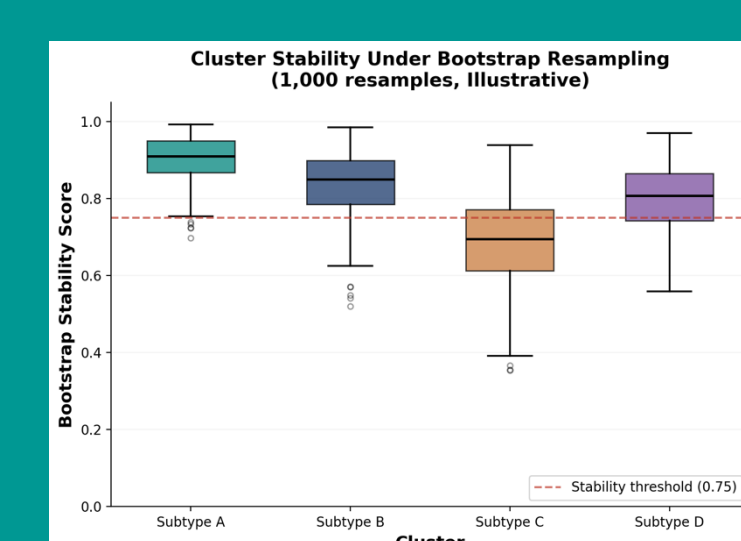
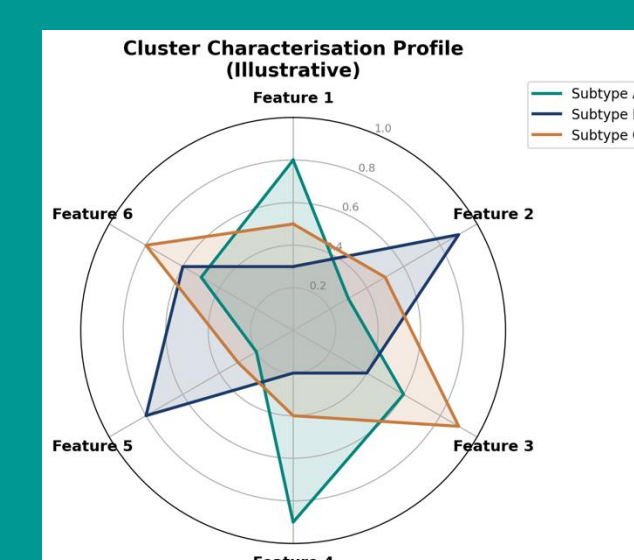
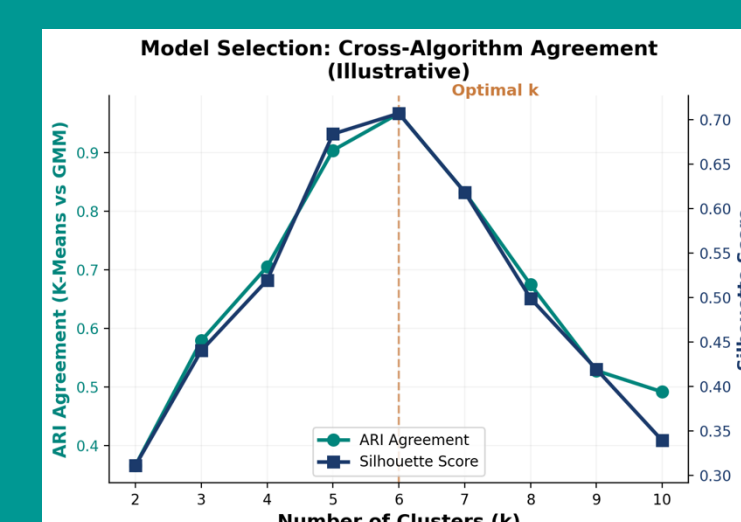
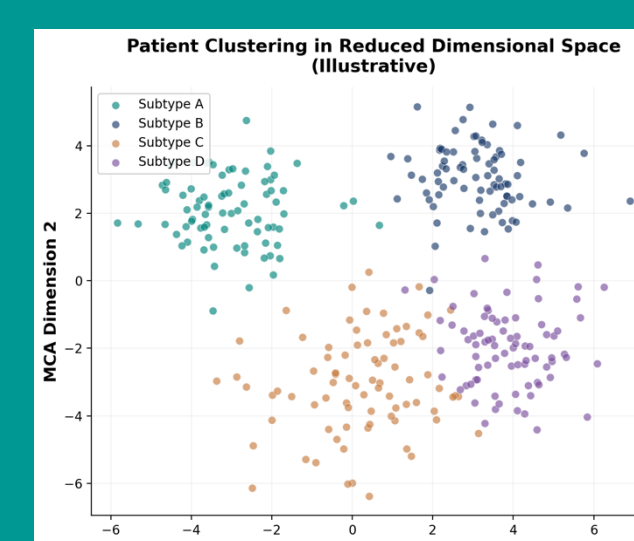


PROJECT 1 - Quality Control & Differential Expression Analysis

- Engineered an end-to-end QC pipeline for high-dimensional biomedical data
- PCA** outlier detection and **KNN/Random Forest** imputation to handle missingness and noise in biological measurements prior to downstream analysis
- Longitudinal statistical modelling across sample cohorts using **regression, hierarchical clustering, and Jaccard similarity**
- Hypothesis testing: **t-tests, chi-squared tests**, and multiple testing correction to identify statistically significant differentially expressed proteins
- Volcano plots** (left) visualise this analysis: each point represents a protein, plotted by statistical significance against magnitude of change, with the most biologically relevant findings appearing in the plot's upper corners

PROJECT 2 - Subtype Discovery via Unsupervised Learning

- Applying unsupervised machine learning to a large-scale dataset (550k+ samples, 18 relational tables) to identify clinically meaningful subgroups
- Applied Multiple Correspondence Analysis (MCA) for dimensionality reduction, followed by three clustering algorithms — K-Means, Gaussian Mixture Models, and HDBSCAN
- Designed a systematic model selection framework using Adjusted Rand Index (ARI) to measure agreement between clustering algorithms, with BIC and silhouette scoring for cluster number
- Validated cluster stability through bootstrap resampling, and cross-referenced results against HDBSCAN's density-based approach to ensure findings were not an artefact of any single method's assumptions
- Executed on GPU-accelerated HPC infrastructure to handle the scale of the dataset



HOW UNIVERSITY UNITS PREPARED ME FOR MSD

CM22009 (Machine Learning) — taught the mathematical rigour behind model selection and justification that underpins every analysis I now run on placement.
CM22016 (Foundations & Frontiers of ML) — trained me to efficiently read and extract insight from AI research papers, a skill I use directly for my IP-led project.
CM12003 (Programming 1) & CM22007 (Software Engineering) — gave me the Python foundations and industry team-working skills I relied on daily at work.

SKILLS GAINED

Practical Skills — Bioinformatics, Machine Learning

Structuring ML pipelines from feature engineering to validation and characterisation, unsupervised learning (K-Means, GMM, HDBSCAN), model selection using cross-algorithm agreement, bootstrap resampling

Theoretical Skills — Mathematics & Statistics

Core statistical testing (t-tests, p-values chi-squared, regression with covariates, multiple testing correction), dimensionality reduction using MCA, constructing and reading advanced plots — volcano plots, PCA plots, enrichment heatmaps, and spider plots

Specialised Knowledge — Neuroscience & Biology

Core molecular biology — the central dogma, proteomics, transcriptomics, genomics, and cell sequencing, practical clinical distinctions like plasma vs. CSF sample types, patient phenotyping, Alzheimer's Disease knowledge

TECHNOLOGIES USED

